

Archel Taneka Sutanto

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SUMMARY

Product Data Scientist with 2+ years of experience using experimentation, causal inference, and machine learning to improve product performance and business outcomes. Delivered AUD 12M+ (IDR 149B+) in annual business impact across conversion, personalisation, and marketplace optimisation initiatives at one of Southeast Asia’s largest online travel platforms. Skilled at partnering with product, engineering, and commercial teams to turn data into product strategy and measurable growth. Recently completed a Master of Data Science at Monash University and available for immediate internship, graduate, or full-time opportunities.

WORK EXPERIENCE

Associate Data Scientist · [tiket.com, ID](#)

Oct 2022 – Jul 2024

One of Southeast Asia's largest online travel platforms serving millions of customers offering app-based platform for booking across flights, hotels, rail, and attractions.

- Increased hotel booking conversion by 2.3%, generating approximately AUD 3.76M (IDR 47B) annual GBV, by deploying a computer vision system that automatically identified low-quality property images across 15,000+ hotel listings and improved listing quality at scale.
- Generated approximately AUD 7.52M (IDR 94B) incremental GBV through a personalised payment recommendation system that analysed behavioural patterns across 15+ payment methods and four customer segments, with impact validated through controlled A/B experimentation.
- Reduced duplicate hotel inventory by 34% and cut manual operational review effort by 87% (120 hours to 15 hours weekly) by developing a CatBoost-based entity matching solution covering 12,000+ hotel properties.
- Delivered approximately AUD 640K (IDR 8B) incremental GBV by identifying high-value personalisation opportunities through recommendation-system experimentation and segmented traveller behaviour analysis, directly influencing roadmap prioritisation for hotel discovery products.
- Saved AUD ~25K (IDR 315M) annually in operational costs by deploying an LSTM-based email prediction system that takes customer reschedule emails and extracts their flight information such as airport names, flight number, boarding time, arrival time, and departure time.

PROJECTS

Causal Inference & Uplift Modeling Platform · [Independent Research](#)

Mar 2026 – May 2026

- Built an end-to-end experimentation and causal inference framework comparing Propensity Score Matching, Double Machine Learning, and uplift modelling approaches for estimating heterogeneous treatment effects from observational data.
- Developed reproducible analytical pipelines using Python, EconML, and CausalML to estimate ATE and CATE, enabling comparison of treatment effectiveness across customer segments rather than relying solely on average outcomes.
- Created an interactive Streamlit application that translated complex causal inference outputs into business recommendations suitable for product and commercial stakeholders.

Slot Filling & Intent Detection · [Monash University](#)

Sep 2025 – Oct 2025

- Achieved 95%+ intent classification accuracy and 80%+ slot F1 score across 21 intents and 121 slot labels by benchmarking and optimising multiple NLP architectures, including JointBiLSTM, JointBiLSTM-Attention, and JointBERT.
- Evaluated model trade-offs across accuracy, complexity, and inference performance to identify the most practical architecture for production deployment.

EDUCATION

M.Sc. Data Science · Monash University, AU <i>(Completed Jun 2026, expected graduation Oct 2026)</i>	2024 – 2026
B.Sc. (Hons.) Computer Science · University of Nottingham, UK <i>(Dual degree with Bina Nusantara University)</i>	2019 – 2020
B.Sc. Computer Science · Bina Nusantara International, ID	2016 – 2020

TECHNICAL SKILLS

Languages: Python, SQL, R
Machine Learning: Scikit-learn, PyTorch, TensorFlow, XGBoost, CatBoost, EconML, CausalML
Experimentation & Analytics: A/B Testing, Causal Inference, Uplift Modelling, Tableau, Power BI
Data & Cloud: PySpark, Databricks, BigQuery, GCP, Docker, MongoDB